CO3020 - Computer Organization 計算機組織

Computer Abstractions and Technology

電腦抽象化與科技

江育綺

2026-09-23



NCU Communication Engineering

Previous Task - CPU Performance

Look at the two CPUs. **Which one finishes** the same program **first**?

CPU A

3.0 GHz

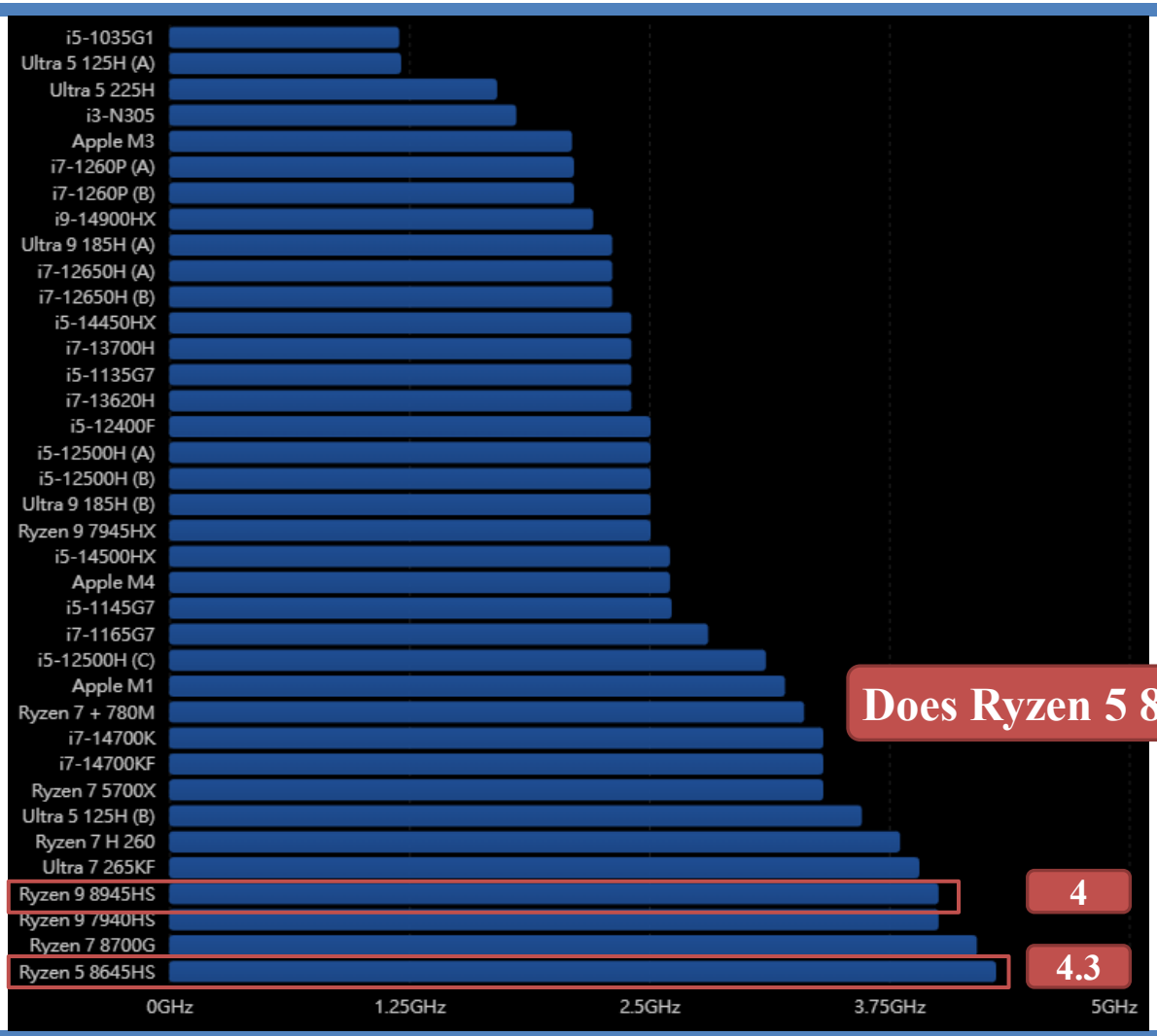
6 billion clock cycles

CPU B

4.0 GHz

10 billion clock cycles

Reported CPU Clock Rates



Learning Goals

- Evaluate CPU performance using execution time and the CPU performance equation.
- Explain how instruction count, CPI, and clock rate affect CPU performance.
- Evaluate power consumption and performance-per-watt under different workloads.
- Analyze performance improvement and its limitations using Amdahl's Law.

Defining Performance

Which airplane has the best performance?





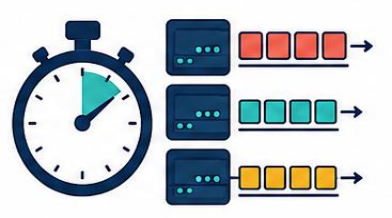




Response Time and Throughput

Response/Execution time

1 task

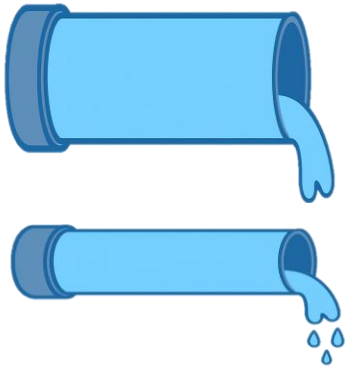


How long it takes to do a task

A shorter response time means **higher** performance.

Throughput

per unit time



Larger throughput

Smaller throughput

Total work done per unit time

A larger throughput means **higher** performance.

Relative Performance

Define Performance = 1/Execution Time

"X is *n* time faster than Y"

$$\text{Performance}_X / \text{Performance}_Y$$
$$= \text{Execution time}_Y / \text{Execution time}_X = n$$

- Example: time taken to run a program

Program A

10s

Program B

15s

Execution Time_B / Execution Time_A
$= 15s \,/\, 10s \;=\; 1.5$

A is **1.5 times** faster than B

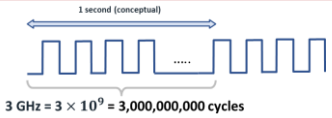
CPU Performance



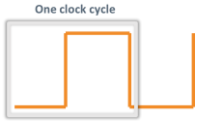
CPU Performance

$$\frac{1}{\text{Execution Time}}$$

- **Clock Rate**



- **Clock Period**



- **Instruction Count**



- **Cycles Per Instruction**



Faster means higher performance

Compare two CPUs

Same program, input, and workload

If Execution Time A < Execution Time B

Performance A > Performance B

Performance

1 / Time

Performance = 1 / Execution time



Check Same program, CPU A: 4 s. CPU B: 6 s. Which is faster? **CPU A**

Measuring Execution Time

Elapsed time includes more than CPU work

Elapsed time

Total response time for the job

CPU execution, OS activities, I/O and waiting.

Measures overall system response

CPU time

Time spent processing a given job

User CPU time + system CPU time

Excludes I/O waits and other jobs' time

One program's timeline

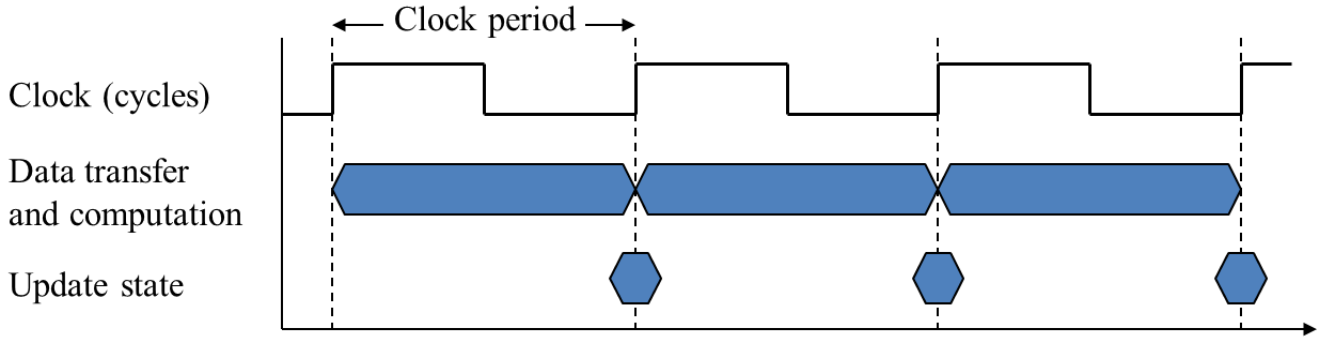


The same elapsed time can contain different amounts of CPU work.

Check A program takes 10 s elapsed time and 6 s CPU time. What fills the other 4 s? **I/O**

Clock Relationships

Operation of digital hardware governed by a constant-rate clock



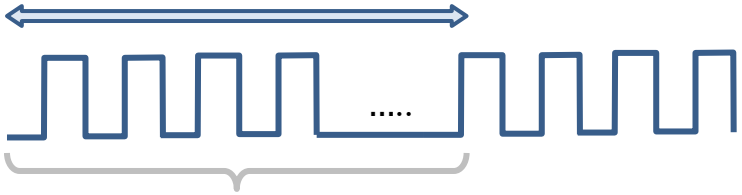
Clock rate cycles per second

3 GHz 3 billion cycles per second

Clock period duration of a clock cycle

1/3 ns 0.333 nanosecond

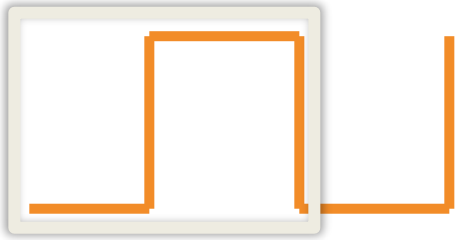
1 second (conceptual)



$$3\ \text{GHz} = 3 \times 10^9 = 3{,}000{,}000{,}000\ \text{cycles}$$
G M K

Clock period =
1 / Clock rate

One clock cycle



$$\frac{1}{3{,}000{,}000{,}000} = \frac{1}{3} \times 10^{-9}$$

CPU Time Calculation

c = a + b
➔ Load a
➔ Load b
➔ Add
➔ Store c

CPU Time = Clock Cycles × Clock period
= IC × CPI × Clock period

The equation links software, instruction behavior, and hardware timing.

Expand the cycle count

Average cycles per instruction

Clock Cycles = Instruction count (IC) × Cycles Per Instruction (CPI)

Instruction count is determined by program, ISA and compiler.

CPU Time

= Clock Cycles × Clock Period

= Clock Cycles ÷ Clock rate



CPU Time Example

Computer A

2 GHz
10s CPU time

Computer B

1.2x Clock cycles of A
6s CPU time

- How fast the computer B clock should be?

$$\text{CPU time} = \text{Clock Cycles} \times \text{Clock Period} = \frac{\text{Clock Cycles}}{\text{Clock Rates}}$$

$$\text{Clock Rate}_B = \frac{\text{Clock Cycles}_B}{\text{CPU Times}_B} = \frac{1.2 \times \text{Clock Cycles}_A}{6s}$$

$$\text{Clock Cycles}_A = \text{CPU Time}_A \times \text{Clock Rate}_A$$
$$= 10\text{s} \times 2\text{GHz} = 20 \times 10^9$$

➔ $$\text{Clock Rate}_B = \frac{1.2 \times 20 \times 10^9}{6s} = \frac{24 \times 10^9}{6s} = 4 \text{ GHz}$$

CPI Example

Computer A

Clock Period 250 ps

CPI 2

Computer B

Clock Period 500 ps

CPI 1.2

- Which one is faster, and by how much?

CPI Example

Computer A

Clock Period 250 ps

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CPI 1.2

- Which one is faster, and by how much?

$$\text{CPU Time}_A = \text{Instruction Count} \times \text{CPI}_A \times \text{Clock Period}_A$$
$$= IC \times 2.0 \times 250ps = IC \times 500ps$$

A is faster

$$\text{CPU Time}_B = \text{Instruction Count} \times \text{CPI}_B \times \text{Clock Period}_B$$
$$= IC \times 1.2 \times 500ps = IC \times 600ps$$

→ $$\frac{\text{CPU Time}_B}{\text{CPU Time}_A} = \frac{IC \times 600ps}{IC \times 500ps} = 1.2$$

By this much

CPI in more details

If different instruction classes take different numbers of cycles

$$Clock\ Cycles = \sum_{i=1}^{n} (CPI_i \times IC_i)$$

Weighted average CPI

$$CPI = \frac{Clock\ Cycles}{Instriction\ Count} = \sum_{i=1}^{n} \left(CPI_i \times \underbrace{\frac{Instruction\ Count_i}{Instruction\ Count}} \right)$$

Relative frequency

Detail CPI Example

- Compiled code sequences using instructions in classes A, B, and C

Class	A	B	C
CPI	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

Detail CPI Example

- Compiled code sequences using instructions in classes A, B, and C

Class	A	B	C
CPI	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

- Sequence 1: IC = 5
  - Clock Cycles
    = 2×1 + 1×2 + 2×3 = 10
  - Avg. CPI
    = 10 / 5 = 2
- Sequence 2: IC = 6
  - Clock Cycles
    = 4×1 + 1×2 + 1×3 = 9
  - Avg. CPI
    = 9 / 6 = 1.5

Performance Summary

The BIG Picture

$$\text{CPU Time} = \text{IC} \times \text{CPI} \times \text{Clock period}$$

$$\text{CPU Time} = \frac{Instruction}{Program} \times \frac{Clock\ Cycles}{Instruction} \times \frac{Seconds}{Clock\ Cycle}$$

	Instruction Count	CPI	Clock Rate
Program	V	V	
Compiler	V	V	
Instruction Set	V	V	V
Organization		V	V
Technology			V

Example: calculate CPU execution time

CPU Time = IC × CPI × Clock Period = IC × CPI ÷ Clock Rate

4457847

Instruction Count

2×10^9

Average CPI

1.5

Clock Rate

3 GHz

Clock rate = 3 GHz

Cycles = IC × CPI = $2 \times 10^9 \times 1.5 = 3 \times 10^9$

CPU time = IC × CPI ÷ Clock Rate

$= 3 \times 10^9 \div 3 \times 10^9 = 1$ s

***Unit check:**

$$instructions \times \frac{cycles}{instruction} \div \frac{cycles}{seconds} = seconds$$

An improvement may change more than one factor.

Slido What is the CPU time? 1 sec

Example: What is the new CPU time?

CPU Time = IC × CPI × Clock Period = IC × CPI ÷ Clock Rate

4457847

Instruction Count

20% less

Average CPI

Stay the same

Clock Rate

Stay the same

Original CPU execution time = 1.0 s

Enter the new time in Slido.

Slido New CPU time = ? **0.8 s**

Example: Speed up CPU Performance

Instruction Count

20% less

Average CPI

Stay the same

Clock Rate

Stay the same

Speed Up Performance by ? %

4457847

Old CPU execution time = 1.0 s

New CPU execution time = 0.8 s

Enter the percentage in Slido.

Distinguish **Speed Up by ? %** **125%**

Padlet Diagnosis: Relationship Description

CPU Performance Relationship

Use your own description

Answer the corresponding question for your group number

Padlet

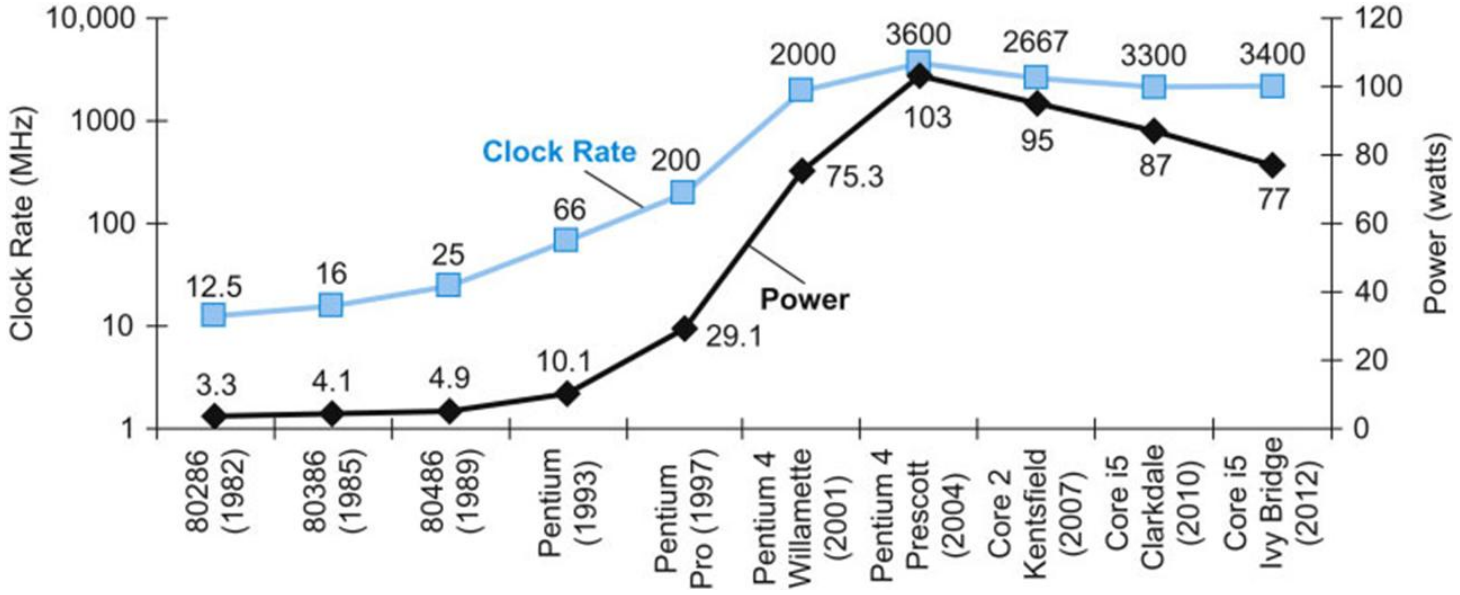
Group 1. **What are the four factors that affect CPU Performance? The definition should be explained.**

Group 2. **Explain the relationship between Clock rate and period. How do they affect CPU performance?**

Group 3. **Explain the relationship between IC and CPI. How do they affect CPU performance?**

Explain Answer the question in the sandbox. **Padlet**

Power Trends



In CMOS IC technology

$$\text{Power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

×30

5V → 1V

×1000

Reducing Power

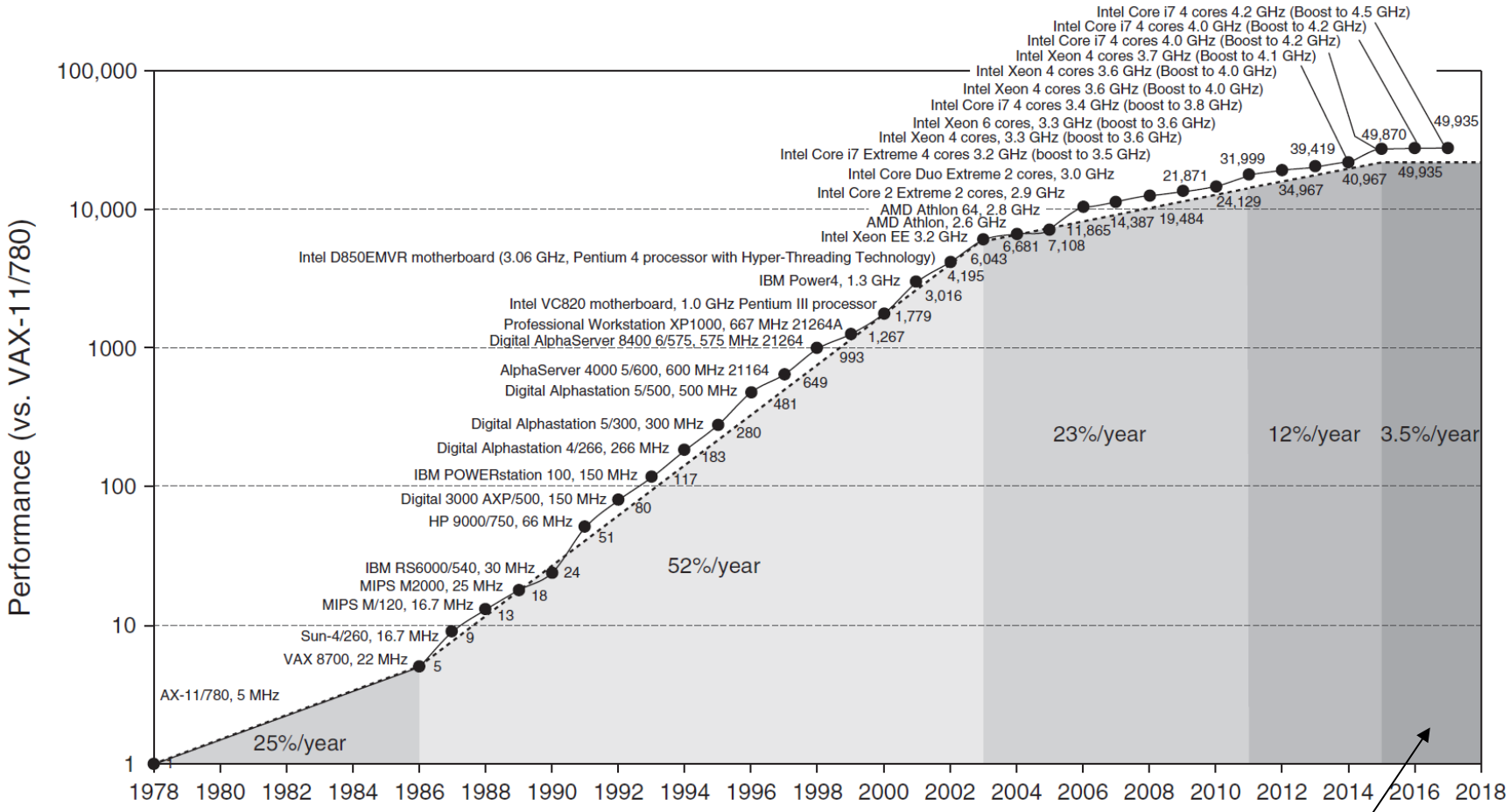
$$\text{Power} = \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency}$$

- Suppose a new CPU has
  - 85% of capacitive load of old CPU
  - 15% voltage and 15% frequency reduction

$$\frac{P_{new}}{P_{old}} = \frac{C_{old} \times 0.85 \times (V_{old} \times 0.85)^2 \times F_{old} \times 0.85}{C_{old} \times V_{old}^{\ 2} \times F_{old}} = 0.85^4 = 0.52$$

- The power wall $P \propto C \times V^2 \times f$
  - We can't reduce voltage further — **Voltage margin for 0 and 1**
  - We can't remove more heat — **increasingly difficult and costly to remove the additional heat**
- How else can we improve performance?

Uniprocessor Performance



Constrained by power, instruction-level parallelism, memory latency

Multiprocessors

- Multicore microprocessors
  - More than one processor per chip
- Requires explicitly parallel programming
  - Compare with instruction level parallelism (ILP)
    - Hardware executes multiple instructions at once
    - Hidden from the programmer
  - Hard to do
    - Programming for performance
    - Load balancing
    - Optimizing communication and synchronization

ILP:	**one core execute multiple instructions at the same time**
Multicore:	**multiple cores execute different tasks at the same time**

SPEC CPU Benchmark

- Programs used to measure performance (execution time)
  - Supposedly typical of actual workload
- Standard Performance Evaluation Corp (SPEC)
  - Develops benchmarks for CPU, I/O, Web, …
- SPEC CPU2006
  - Elapsed time to execute a selection of programs
    - Negligible I/O, so focuses on CPU performance
  - Normalize relative to reference machine
  - Summarize as **geometric mean** of performance ratios
    - CINT2006 (integer) and CFP2006 (floating-point)

$$\sqrt[n]{\prod_{i=1}^{n} \text{Execution time ratio}_i}$$

SPECspeed 2017 Integer benchmarks on a 1.8 GHz Intel Xeon E5-2650L

Description	*Name*	*Instruction Count x 10^9*	*CPI*	*Clock cycle time (seconds x 10^–9)*	*Execution Time (seconds)*	*Reference Time (seconds)*	*SPECratio*
Perl interpreter	perlbench	2684	0.42	0.556	627	1774	2.83
GNU C compiler	gcc	2322	0.67	0.556	863	3976	4.61
Route planning	mcf	1786	1.22	0.556	1215	4721	3.89
Discrete Event simulation - computer network	omnetpp	1107	0.82	0.556	507	1630	3.21
XML to HTML conversion via XSLT	xalancbmk	1314	0.75	0.556	549	1417	2.58
Video compression	x264	4488	0.32	0.556	813	1763	2.17
Artificial Intelligence: alpha-beta tree search (Chess)	deepsjeng	2216	0.57	0.556	698	1432	2.05
Artificial Intelligence: Monte Carlo tree search (Go)	leela	2236	0.79	0.556	987	1703	1.73
Artificial Intelligence: recursive solution generator (Sudoku)	exchange2	6683	0.46	0.556	1718	2939	1.71
General data compression	xz	8533	1.32	0.556	6290	6182	0.98
Geometric mean							2.36

SPEC Power Benchmark

- Power consumption of server at different workload levels (10% increase each run)
  - Performance: ssj_ops/sec
  - Power: Watts (Joules/sec)

$$\text{Overall ssj_ops per Watt} = \left(\sum_{i=0}^{10} \text{ssj_ops}_i\right) \Big/ \left(\sum_{i=0}^{10} \text{power}_i\right)$$

If the workload is only 10%, the server to consume only 10% of its full-load power?

SPECpower_ssj2008 for Xeon E5-2650L

Target Load %	Performance (ssj_ops)	Average Power (watts)
100%	4,864,136	347
90%	4,389,196	312
80%	3,905,724	278
70%	3,418,737	241
60%	2,925,811	212
50%	2,439,017	183
40%	1,951,394	160
30%	1,461,411	141
20%	974,045	128
10%	485,973	115
0%	0	48
Overall Sum	26,815,444	2,165
∑ssj_ops / ∑power =		12,385

Pitfall: Amdahl's Law

- Improving an aspect of a computer and expecting a proportional improvement in overall performance

$$T_{improved} = \frac{T_{affected}}{improvement\ factor} + T_{unaffected}$$

- Example: multiply accounts for 80s/100s
  - How much improvement in multiply performance to get 5× overall?

$$\frac{\mathbf{100}}{\mathbf{5}} = \frac{80}{n} + 20$$

  - Can't be done!

- Corollary: **make the common case fast**

Fallacy: Low Power at Idle

- Look back at i7 power benchmark
  - At 100% load: 258W
  - At 50% load: 170W (66%)
  - At 10% load: 121W (47%)
- Google data center
  - Mostly operates at 10% – 50% load
  - At 100% load less than 1% of the time
- Consider designing processors to make power proportional to load

Pitfall: MIPS as a Performance Metric

- MIPS: Millions of Instructions Per Second
  - Doesn't account for
    - Differences in ISAs between computers
    - Differences in complexity between instructions

$$\text{MIPS} = \frac{\text{Instruction count}}{\text{Execution time} \times 10^6}$$

$$= \frac{\text{Instruction count}}{\frac{\text{Instruction count} \times \text{CPI}}{\text{Clock rate}} \times 10^6} = \frac{\text{Clock rate}}{\text{CPI} \times 10^6}$$

- CPI varies between programs on a given CPU

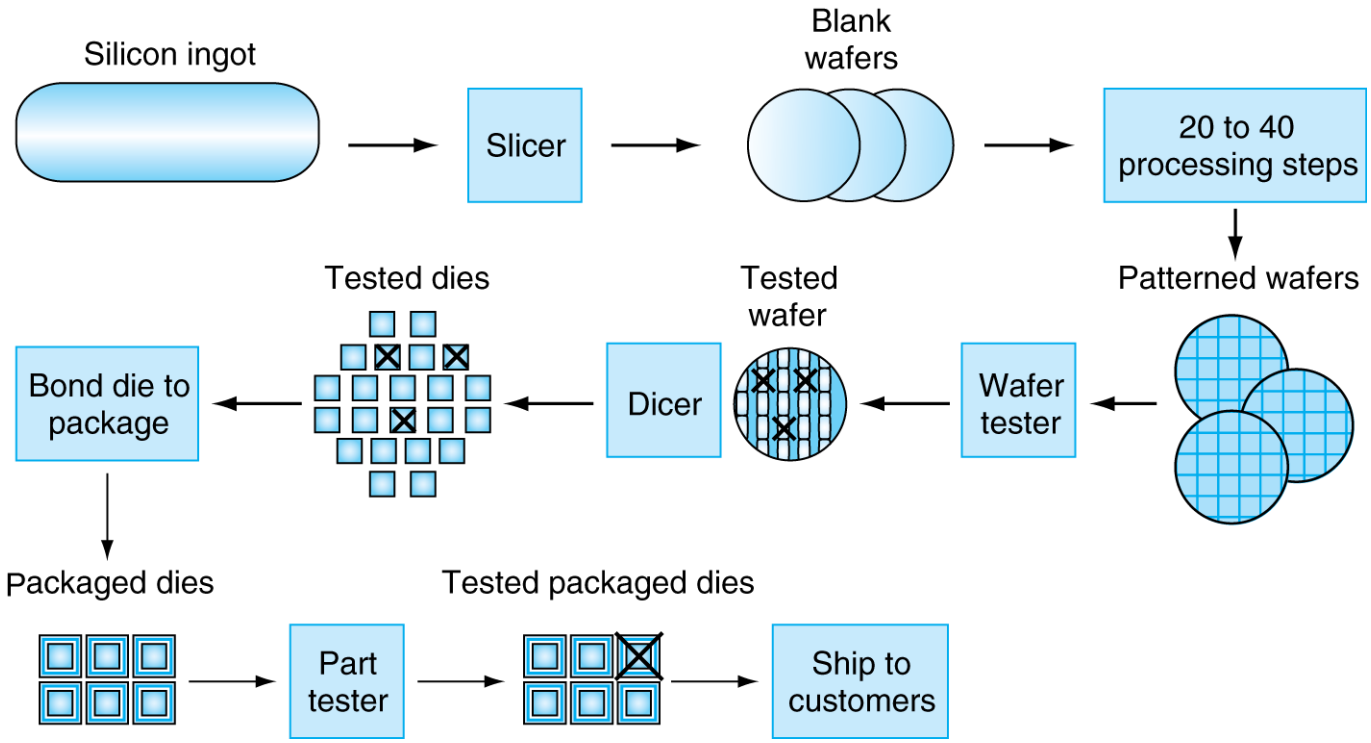
Pitfall: MIPS as a Performance Metric

- What's the MIPS for these two computers?
- Which computer is faster?

$$\text{MIPS} = \frac{\text{Clock rate}}{\text{CPI} \times 10^6}$$

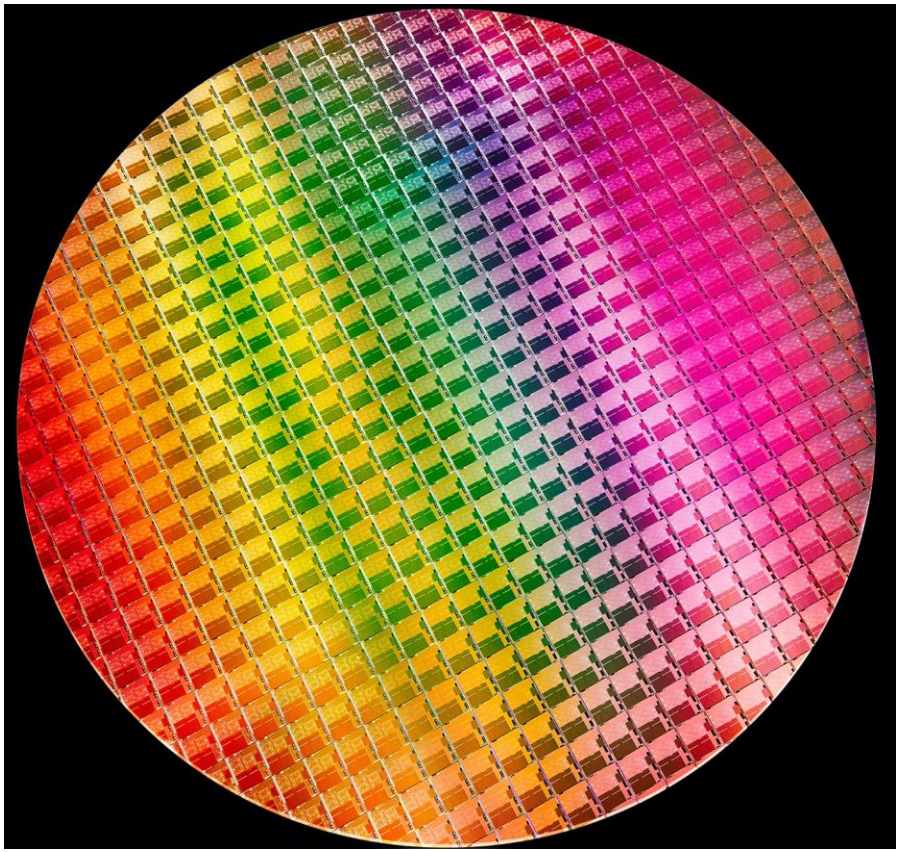
Measurement	Computer A	Computer B
Instruction Count	10 billion	8 billion
Clock rate	4 GHz	4 GHz
CPI	1.0	1.1
MIPS	4×10^3	3.63×10^3
Faster?	2.5 sec	2.2 sec

Manufacturing ICs



- Yield: proportion of working dies per wafer

Intel® Core 10th Gen



- 300mm wafer, 506 chips, 10nm technology
- Each chip is 11.4 x 10.7 mm

Integrated Circuit Cost

$$\text{Cost per die} = \frac{\text{Cost per wafer}}{\text{Dies per wafer} \times \text{Yield}}$$

$$\text{Dies per wafer} \approx \text{Wafer area} / \text{Die area}$$

$$\text{Yield} = \frac{1}{(1 + (\text{Defects per area} \times \text{Die area}/2))^2}$$

- **Nonlinear relation** to area and defect rate
  - Wafer cost and area are fixed
  - Defect rate determined by manufacturing process
  - Die area determined by architecture and circuit design

Concluding Remarks

- Cost/performance is improving
  - Due to underlying technology development
- Hierarchical layers of abstraction
  - In both hardware and software
- Instruction set architecture
  - The hardware/software interface
- Execution time: the best performance measure
- Power is a limiting factor
  - Use parallelism to improve performance